

Linking 3-D soil macropores and root architecture to near saturated hydraulic conductivity of typical meadow soil types in the Qinghai Lake Watershed, northeastern Qinghai–Tibet Plateau

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ABSTRACT

The importance of soil macropores has been widely recognized, which permit the rapid flow of water through soils. However, few studies have focused on the soil macropores of different alpine meadow soil in cold environments in the Qinghai–Tibet Plateau. This study aims to use computed tomography to quantify the soil macropore and root architecture of *Meadow swamp* soil, *Sub-alpine shrub meadow* soil, *Sub-alpine grassland meadow* soil and *Sub-alpine meadow* soil in the Qinghai Lake Watershed of the northeastern Qinghai–Tibet Plateau. A total of twelve soil cores (0–50 cm deep) were collected at four typical soil types under alpine meadow, with three replicates. At each site, the collected cores were scanned using the X-ray 3D microscope (Nano Voxel-2000, Sanying Precision Instruments Co., Ltd.). The soil macropores and root architecture of the soil cores were explored using the X-ray computed tomography method. The results indicated that the *Meadow swamp* soil and *Sub-alpine shrub meadow* soil had higher macropores than *Sub-alpine grassland meadow* soil and *Sub-alpine meadow* soil. *Sub-alpine grassland meadow* soil and *Sub-alpine meadow* soil had macropores that were mainly concentrated in the 0–100-mm soil depth. Moreover, macropore networks of *Meadow swamp* soil and *Sub-alpine shrub meadow* soil were more convoluted and continuous than those of *Sub-alpine grassland meadow* soil and *Sub-alpine meadow* soil. Root density was greater in the *Meadow swamp* soil and *Sub-alpine shrub meadow* soil than those in the *Sub-alpine meadow* soil and *Sub-alpine grassland meadow* soil. The soil macropore architecture with a high continuity could facilitate greater water flow to deeper soil layers in the *Meadow swamp* soil and *Sub-alpine shrub meadow* soil than in the *Sub-alpine grassland meadow* soil and *Sub-alpine meadow* soil. Root density, root node density, length density and surface area density and branch density were important parameters influencing soil macropore. Soil macropore length density was significantly and positively correlated to soil near saturated hydraulic conductivity $K(10)$. Soil water status and alpine shrub root induce greater macroporosity of *Meadow swamp* soil and *Sub-alpine shrub meadow* soil than *Sub-alpine grassland meadow* soil and *Sub-alpine meadow* soil, and thus present a greater likelihood of larger macropores.

1. Introduction

Soil macropores are large voids in the soil that consist of worm holes, root paths, soil cracks, and inter-aggregate voids, and they permit rapid flow of water through soils (Lin et al., 2005; Katuwal et al., 2015; Jarvis, 2007). The importance of macropores has been widely

recognized, which was the preferential pathways for water, air, and chemicals in soils (Beven and Germann, 1982; Lin et al., 2005; Jarvis, 2007). The impact of soil macropores on water transport depends mainly on their arrangement and three-dimensional (3-D) characteristics, including: length, diameter, volume, surface area, mean tortuosity, mean angle, and density (Bastardie et al., 2003; Luo et al., 2008;

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Perret et al., 2000; Pierret et al., 2002). Soil type is one of the important factors which influences soil structure (Gantzer and Anderson, 2002; Udawatta et al., 2008; Mooney and Morris, 2008). However, knowledge gaps still exist about how soil macropores change among different soil types under alpine meadow vegetation in high altitude and cold environments.

X-ray computed tomography (CT) has been used in recent years as a new method to quantify soil pores at a much higher resolution than the previous methods, such as dye tracing, spectral image analysis and soil thin section (Katuwal et al., 2015; Larsbo et al., 2014; Vogel et al., 2010). Li et al. (2016) quantified the macropore characteristics of loess soil under two perennial plants using a hospital CT scanner. Meng et al. (2017) quantified soil macropore characteristics in different forest communities using an industrial CT scanner in a mountain area of North China. Starkloff et al. (2017) quantified the influence of freezing-thawing cycles on the pore network of a silty clay loam and a loamy sand soil using CT. Recently, many studies have been conducted using X-ray CT to observe plant roots in situ. Koebernick et al. (2018) visualized and quantified root-soil contact using X-ray CT. Hu et al., 2019a,b quantified the effects of exclosures on CT-measured soil macropore and root characteristics beneath the encroached shrub *C. microphylla* L. in the Inner Mongolia grassland. The above mentioned studies indicate that X-ray CT is an appropriate method to quantify the root architecture and soil structure in undisturbed soil cores.

Alpine soils develop in high-latitude areas under unique and harsh climate conditions characterized by low temperatures, aridity, ultraviolet radiation, freeze-thaw cycles and short growing seasons. In the cold climates of high-altitude alpine regions, soil and ecosystem formation are limited by temperature, parent materials, and nutrients because of subordinate chemical weathering conditions. Because of the extreme environmental conditions of alpine mountainous regions, alpine ecosystems are vulnerable to climate change and human disturbance. In high-latitude areas, the soil structures of different soil types may vary from those in low-latitude areas. Moreover, Schwärzel et al. (2012) indicated that root would induced preferential flow which would trigger rapid lateral subsurface flow. Hu et al. (2015) attributed high numbers of macropores in soil to greater root development and subsequent root decay. Hu et al. (2019a,b) quantified soil macropores and root architecture through applying X-ray CT to soils under encroaching *C. microphylla* L. in the shrub-encroached grasslands of Inner Mongolia. Up to now, researches on the direct linkages between soil macropores, root architecture and hydrological attributes for the alpine meadow soils in the Qinghai-Tibet Plateau are lacking. Therefore, the objective of this study was to quantify the soil macropore and root architecture of typical meadow soil types by X-ray computed tomography (CT) to reveal the interaction between soil macropore and root architecture and hydraulic conductivity in the Qinghai Lake watershed, northeastern Qinghai-Tibet Plateau.

2. Materials and methods

2.1. Study sites and core samples

This study was performed within the Qinghai Lake Watershed along the northeastern Qinghai-Tibet Plateau, which is in a semiarid, cold, high-altitude climatic area (Fig. 1). The mean annual temperature was 0.1 °C, and precipitation was 400 mm in the watershed between 1957 and 2015, respectively. Approximately 70–80% of the annual precipitation occurs during summer and autumn. Winters are clear, dry, and cold, and summers are warm and wet.

We selected the sampling areas based on four soil types under alpine meadow vegetation with different plant composition and soil water conditions (Meadow swamp soil, Sub-alpine shrub meadow soil, Sub-alpine grassland meadow soil and Sub-alpine meadow soil) (Fig. 1), which are equivalent to Gleysols, Matti-Gelic Cambosols, Matti-Gelic Cambosols and Matti-Gelic Cambosols according to the USDA Soil Taxonomy (Soil

Survey Staff, 1987). In addition, the soil profiles of the four soil types are shown in Fig. 1. Twelve intact soil cores were collected at the four soil type sites, with 3 replicates per site. Plexiglas cylinders (4-mm-thick walls, 10-cm diameter and 50-cm length) were used to house the soil cores. Core extraction followed the previously described procedure (Hu et al., 2015, 2016).

Soil samples were collected at 0–50 cm for all the soil types. Sand fraction (2000–50 µm) was measured by sieving, and the silt (50–2 µm) and clay (< 2 µm) fractions were determined using the pipette method. Organic carbon content was determined by the method of Walkley and Black (1934) and converted to organic matter content by multiplying by 1.724. The soil characteristics at 0–50 cm of different soil types are listed in Table 1.

Hydraulic conductivity is a measure of the ability of a soil to transmit water when submitted to a hydraulic gradient, and controls water fluxes through the landscape (Lebron et al., 2007). The hydraulic conductivity can then be used to evaluate the infiltration capacity (Li et al., 2008). In this study, a minidisk infiltrometer (Decagon Devices, 2012) was used to measure the soil near saturated hydraulic conductivity $K(h)$ with pressure heads of 10, 30, and 60 mm ($K(10)$, $K(30)$ and $K(60)$) at each core.

2.2. CT scanning and image analysis

A Nano Voxel-2000 X-ray 3-D microscope (Sanying Precision Instruments Co., Ltd., China), with a 20× lens detector, was used to scan soil cores at an energy level of 80 kV and 95 µA. The scanning produced images with 1536 × 1932 pixels per slice; thus, the resolution of voxel was 1700 × 1700 × 1500 in each reconstructed image, which resulted in a maximum resolution of 75 µm. And, the FDK algorithm was used for this rescaling (Zhang et al., 2018).

The thresholds was determined by a plexiglas cylinder with a known diameter measured by a digital caliper, which was regarded as an artificial macropore. And, the plexiglas cylinder was inserted into an undisturbed soil core and then pulled out. We selected the upper, middle and lower parts of the artificial macropore. First, we set a threshold for the threshold sample and use Avizo 9.0 (FEI, 2016) to calculate the macropore size based on the image analysis and compared it with the actual size. If the difference between the calculated result and the actual size was too large, we set another threshold until the difference drops below 1% (Li et al., 2013). And for all scanned samples, the same threshold determined by described procedure was used.

Slices were prepared following the procedure of Luo et al. (2010) before segmentation. The slice images were reconstructed and analysed using AVIZO Fire 9.0 (FEI, America). A cylindrical cropping tool was applied to obtain a region of interest (ROI) to remove obvious disturbances caused by the sampling. After segmentation, the macropore networks were reconstructed and visualized using Avizo 9.0 (FEI 2016). The macropore density, macropore distribution along the column depth, macropore volume, macropore size distribution (sorted by macropore volume), mean macropore volume, and total surface area were calculated using Avizo 9.0 (FEI 2016). Since soil macropores are three-dimensional (3-D), skeletonization of macropores is necessary to accurately quantify the topology and length and to effectively visualize the networks in three dimensions. The skeletons of macropore networks were generated by determining the nearest distance to the border voxels and thinning based on the nearest distance using Avizo 9 (FEI 2016).

After skeletonization, macropore number density ($g_d(m^{-3})$) was defined as the number of macropore networks ($N_m(-)$) in a unit volume ($V(m^{-3})$):

$$\rho_d = \frac{N_m}{V} \quad (1)$$

A node refers to an intersection at which two macropore branches are connected. The number of nodes in each macropore network was



Fig. 1. Soil profiles of different soil types (Meadow Swamp soil, Sub-alpine shrub meadow soil, Sub-alpine meadow soil, Sub-alpine grassland meadow soil).

Table 1

Properties of the four soil types in the study site.

Soil types	Vegetation	Soil depth (cm)	Organic matter (g kg ⁻¹)	Total phosphorus (g kg ⁻¹)	Total nitrogen (g kg ⁻¹)	Particle size composition (%)			Bulk density (g cm ⁻³)
						Clay (%)	Silt (%)	Sand (%)	
Meadow Swamp soil	<i>Kobresia</i> meadow	0–10	28.17	0.18	0.021	10.59	45.74	43.68	1.06
		10–20	20.25	0.25	0.020	4.64	19.69	75.67	1.00
Sub-alpine shrub meadow soil	<i>Potentilla fruticosa</i> shrub, <i>Kobresia</i> meadow	0–10	165.37	0.0330	0.7187	63.85	27.63	8.52	0.57
		10–20	174.90	0.0188	0.6465	47.00	44.26	8.73	0.70
		20–40	61.10	0.3471	0.0196	46.22	45.37	8.42	0.75
Sub-alpine meadow soil	<i>Kobresia</i> meadow	0–10	203.00	0.0279	1.0888	76.53	19.34	4.14	0.48
		10–20	216.00	0.0200	0.9061	50.43	42.73	6.85	0.49
		20–40	165.00	0.0180	0.8000	70.30	20.00	9.70	0.48
Sub-alpine grassland meadow soil	<i>Kobresia</i> meadow	0–10	55.776	0.018	0.34	6.19	29.37	64.44	1.07
		10–20	36.621	0.018	0.24	11.43	56.67	31.90	1.10
		20–40	25.916	0.016	0.18	8.64	30.08	61.28	1.19

recorded, and we calculated the node density (ρ_n (m⁻³)) as the number of nodes (N_n (-)) in a unit volume (V (m⁻³)):

$$\rho_n = \frac{N_n}{V} \quad (2)$$

The length density (g_l (km m⁻³)) was also analysed as the total actual macropore length (L_t) in a unit volume (V) (Luo et al., 2010).

$$\rho_l = \frac{L_t}{V} \quad (3)$$

The mean tortuosity (τ) was calculated as ratio of the total actual macropore length (L_t) to total straight-line distance (L_p) of all the macropores in a certain volume (Luo et al., 2010). And, the mean tortuosity means the ratio of total pore length to that of the shortest possible path.

$$\bar{\tau} = \frac{\sum_{i=1}^n L_{ti}}{\sum_{i=1}^n L_{pi}} \quad (4)$$

where i is the index of a macropore branch and n is the total number of macropore branches.

The inclination of a macropore branch was characterized by an angle (θ) away from vertical direction. The mean angle was calculated as:

$$\bar{\theta} = \frac{\sum_{i=1}^n L_{ti} \theta_i}{\sum_{i=1}^n L_{ti}} \quad (5)$$

Assuming that all macropores are cylindrical, its mean equivalent hydraulic radius was calculated by the total volume V_t and total length L_t of the macropore:

$$\bar{r} = \sqrt{\frac{V_t}{L_t \pi}} \quad (6)$$

Shrub roots were determined by the growth method and the threshold method (Hu et al., 2019a,b; Kuka et al., 2013). Several materials, including the acrylic cylinder, pores, roots, soil mineral particles, organic matter and microorganisms, existed within each sample. In the intensity histogram of raw data, three peaks could be distinguished: -900 HU for macropores, 0 HU for shrub roots, and 450 for soil material (solid) located on the root peak. The Hounsfield unit between -650 and -550 was obtained from the intensity histogram of the raw data by fitting a parabola, which was similar to the method of Katuwal et al. (2015). The soil macropore was segmented from the reconstructed CT data by using thresholding. The roots were segmented from the reconstructed CT data by using the region growing method (Gregory et al., 2003) in AVIZO Fire 9.0. In brief, an interactive module allowed a human operator to track each root through a stack of CT slices by marking the center of the root region. Therefore, a fixed threshold was used to distinguish the pores, the roots and the solid materials. Because there is some overlap in values of image density for roots, soil and container as a consequence of partial-volume effects, it was necessary to apply some additional connectivity rules to improve root identification. A 3D connectivity routine was used to connect neighboring voxels of

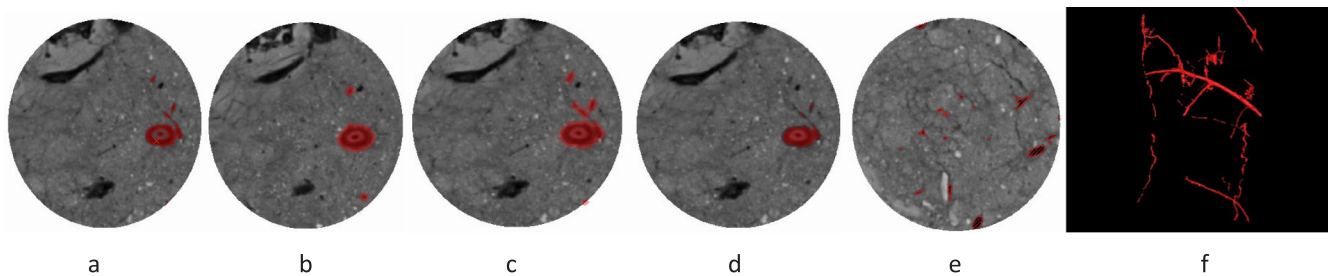


Fig. 2. Procedure used for the root analysis in the study. (a) Selected root space of the sample; (b and c) enlarged root area; (d) operation to shrink all slice voxels; (e) root space of sample; and (f) root networks reconstructed and visualized (Hu et al., 2019a,b).

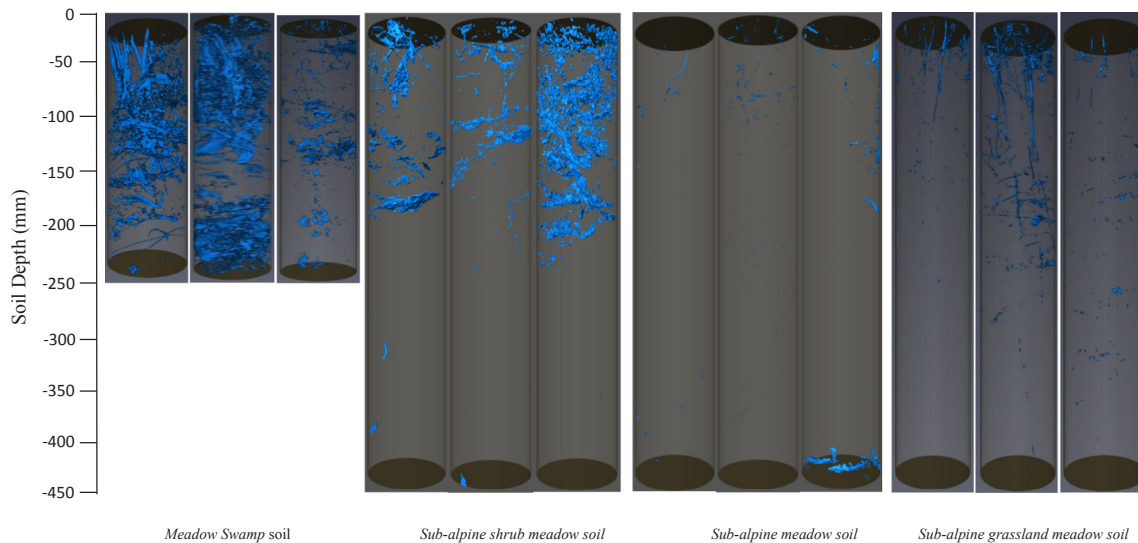


Fig. 3. Three-dimensional visualization of soil macropore networks in the soil columns of the four soil types. Blue denotes macropores, and grey denotes non-pores. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

similar image density starting at the plane corresponding to the base of the seed and working down through each slice of data. The routine identified the center of mass for each root. From these calculations the area of each root slice was recorded, and root length and volume calculated by summing the number and area, respectively, of slices with center of mass identified (Gregory et al., 2003). To avoid selecting too many impurities, we used the magic wand tool (FEL, 2016) to select the root in a two-dimensional (2-D) image and apply the wand tool to all slices, and the shape of the root was shown in the visual window. If the selected root area didn't match the authentic root volume and surface, it was necessary to zoom in on the selected area to ensure that the internal valid voxels were counted as root (Fig. 3a). If the enlarged root area of a 2-D slice included unnecessary area at the edge of the root system (Fig. 3b), all slice voxels were shrunk until the selected parts yielded the authentic root volume (Fig. 3c and d). If the root in the soil core was not continuous, select another root spot in the 2D slice till all the roots were found. After segmentation and binarization, the root networks were visualized. An illustration of the processing steps is presented in Fig. 2. And, the verification of root characteristics from the X-ray micro-CT images followed the procedure of Hu et al. (2019a,b).

2.3. Statistical analyses

All statistical analyses were conducted using SPSS software (version 13.0, SPSS Inc., Chicago, IL). Differences among the measured parameters between the different treatments were analysed using Fisher's protected least significant difference (LSD) test. The formula for the least significant difference is:

$$LSD_{A,B} = t_{0.05/2, DFW} \sqrt{MSW(1/n_A + 1/n_B)} \quad (7)$$

where: t = critical value from the t-distribution table, MSW = mean square within, obtained from the results of the ANOVA test, n = number of scores used to calculate the means.

3. Results

3.1. Visualization of macropore networks

Fig. 3 shows a 3-D visualization of the soil macropores of typical meadow soil types. We found that the spatial characteristics of soil macropore were significantly different among the four typical meadow soils. The number of macropores of the *Meadow swamp* soil and *Sub-alpine shrub meadow* soil was larger than that of *Sub-alpine grassland meadow* soil and *Sub-alpine meadow* soil. The macropore networks in the soil profiles of the *Meadow swamp* soil and *Sub-alpine shrub meadow* soil were more vertically oriented and continuous than those in the soil profiles of *Sub-alpine grassland meadow* soil and *Sub-alpine meadow* soil.

3.2. CT-measured macroporosity

The macropore characteristics of the four soil types are listed in Tables 2. The mean macroporosity of the *Meadow swamp* soil was the largest ($0.0282 \text{ m}^3 \text{ m}^{-3}$), exceeding the mean macroporosity of the *Sub-alpine shrub meadow* soil ($0.0074 \text{ m}^3 \text{ m}^{-3}$), *Sub-alpine grassland meadow* soil ($0.0009 \text{ m}^3 \text{ m}^{-3}$) and *Sub-alpine meadow* soil ($0.0005 \text{ m}^3 \text{ m}^{-3}$) (Table 2). Significant differences were observed among the mean soil macroporosity between *Meadow swamp* soil and *Sub-alpine grassland meadow* soil, *Sub-alpine meadow* soil, and *Sub-alpine shrub meadow* soil.

Table 2
Soil macropore characteristics of all soil columns as determined using the CT images of the four soil types.[†]

	Macropore number density ($\times 10^5$) (no.m ⁻³)	Macroporosity (m ³ .m ⁻³)	Surface area density (m ² .m ⁻³)	Length density (km.m ⁻³)	Node density ($\times 10^5$) (no.m ⁻³)	Branch density (no.m ⁻³)	Tortuosity (-)	Mean angle (°C)	Hydraulic radius(mm)	Mean macropore size (mm ³)	mean Coordination Number(-)
<i>Meadow Swamp</i> soil	42.4804 (± 25.5709)a	0.0282 (± 0.0261)a	0.0663 (± 0.0537)	18.0363 (± 15.2828)a	60.1001 (± 44.961)a	66.5545 (± 55.2808)a	1.6395 (± 0.0499)	60.1227 (± 6.7832)	0.6702 (± 0.0639)	5.4543 (± 3.0315)	2.0663 (± 0.2985)
<i>Sub-alpine shrub meadow</i> soil	6.2667 (± 4.4587)b	0.0074 (± 0.0051)	0.0378 (± 0.0441)	3.2833 (± 0.8188)b	11.7333 (± 3.3115)b	10.6300 (± 2.9905)b	1.0200 (± 0.3101)	58.4033 (± 4.7829)	0.8167 (± 0.2139)a	5.3000 (± 4.5194)	1.7871 (± 0.1609)a
<i>Sub-alpine meadow</i> soil	3.0241 (± 1.0279)b	0.0005 (± 0.0005)b	0.0019 (± 0.0014)	0.4774 (± 0.351)b	2.2346 (± 1.4115)b	1.6619 (± 1.3725)b	0.8978 (± 0.918)	56.82 (± 9.4461)	0.5548 (± 0.0717)b	1.7540 (± 1.6613)	1.3825 (± 0.2875)b
<i>Sub-alpine grassland meadow</i> soil	4.2294 (± 3.4729)b	0.0009 (± 0.0009)b	0.0028 (± 0.0033)	0.7503 (± 0.6895)b	3.2446 (± 2.6479)b	2.3702 (± 2.144)b	0.9216 (± 0.3715)	47.2478 (± 10.4928)	0.581 (± 0.05)b	1.8478 (± 0.6451)	1.4052 (± 0.1857)b

[†] Column for each data group by the same letter (lowercase and uppercase for columns and rows, respectively) are not significantly different at $p = 0.05$, the absence of letters indicates no significant differences for that property. Data in the parentheses are standard deviations.

To investigate the variation of soil macroporosity with depth, macroporosity distribution for each sample was calculated slice per slice. The distributions of macroporosity with sample height (profile depth) for four soil types are presented in Fig. 4. The change in macroporosity with profile depth was significantly different among the four soil types. The curves were wavier for the *Meadow swamp* soil and *Sub-alpine shrub meadow* soil than for *Sub-alpine grassland meadow* soil and *Sub-alpine meadow* soil. For the *Meadow swamp* soil, macroporosity was represented by a wavy curve, which showed a fast decrease at soil layers of 0–100 mm and then remained stable from 100 to 150 mm, and increased at soil depths of 150–250 mm. For the *Sub-alpine shrub meadow* soil, the macroporosity curve showed a rapid decrease at soil layers of 0–200 mm and then remained stable from 200 to 450 mm. For the *Sub-alpine meadow* soil, the macroporosity curve showed a fast decrease at soil layers of 0–100 mm, and then became relatively stable at depths of 100–400 mm. For the *Sub-alpine grassland meadow* soil, the macroporosity was mainly concentrated at soil layers of 0–400 mm, and a rapid decrease at soil layers of 0–50 mm, and then remained stable from 50 to 450 mm.

3.3. Macropore number density, total surface area density, macropore length density, macropore tortuosity, mean angle, and node density

The *Meadow swamp* soil had the largest total surface area density ($0.0663 \text{ m}^2 \text{ m}^{-3}$), length density (18.036 km m^{-3}) and node density ($60.1001 \times 10^5 (\text{m}^{-3})$), whereas the *Sub-alpine shrub meadow* soil had a lower total surface area density ($0.0378 \text{ m}^2 \text{ m}^{-3}$), length density (3.2833 km m^{-3}) and node density ($11.7333 \times 10^5 (\text{m}^{-3})$). The *Sub-alpine grassland meadow* soil had a greater total surface area density ($0.0028 \text{ m}^2 \text{ m}^{-3}$), length density (0.7503 km m^{-3}) and node density ($3.2446 \times 10^5 (\text{m}^{-3})$) than *Sub-alpine grassland meadow* soil ($0.0019 \text{ m}^2 \text{ m}^{-3}$), $0.4774 (\text{km m}^{-3})$ and $2.2346 \times 10^5 (\text{m}^{-3})$.

The mean macropore tortuosities were 1.6395, 1.0200, 0.8978 and 0.9216 for the *Meadow swamp* soil, *Sub-alpine shrub meadow* soil, *Sub-alpine meadow* soil and *Sub-alpine grassland meadow* soil, respectively (Table 2). The highly continuous macropores in the *Meadow swamp* soil and *Sub-alpine shrub meadow* soil had mean tortuosities of 1.5834–1.6788 and 0.8600–1.1800, respectively, while those of the *Sub-alpine meadow* soil and the *Sub-alpine grassland meadow* soil were 0.6079–1.1878 and 0.4900–1.1400, respectively. The tortuosity of soil macropore in the *Sub-alpine meadow* soil and the *Sub-alpine grassland meadow* was lower than the tortuosity of soil macropore in the *Meadow swamp* soil and *Sub-alpine shrub meadow* soil; this occurred because there were fewer observed biopores in the *Sub-alpine meadow* soil and the *Sub-alpine grassland meadow*.

Meadow swamp soil and *Sub-alpine shrub meadow* soil had mean macropore angles of 60.1227° and 58.4033° , respectively, indicating that the macropores were predominantly longitudinal oriented, which was partly related to the large proportion of longitudinal oriented roots. In comparison, *Sub-alpine meadow* soil and *Sub-alpine grassland meadow* soil had a small mean angle of 56.8200° and 47.2478° . Meanwhile, the mean macropore branch density for the *Meadow swamp* soil were 66.5545, and these values were approximately six times than that of the *Sub-alpine shrub meadow* soil (10.6300) (Table 2). Similarly, the mean macropore brach density for the *Sub-alpine grassland meadow* soil was 2.3712, which was higher than that of the *Sub-alpine meadow* soil (1.6619). Similarly, the macropore number densities of the *Meadow swamp* soil ($42.4804 \times 10^5 \text{ m}^{-3}$) and *Sub-alpine shrub meadow* soil ($6.2667 \times 10^5 \text{ m}^{-3}$) were the larger than that of the *Sub-alpine meadow* soil ($3.0241 \times 10^5 \text{ m}^{-3}$) and *Sub-alpine grassland meadow* soil ($4.2294 \times 10^5 \text{ m}^{-3}$).

3.4. Macropore size distribution

Significant differences in macropore size were not observed among the three replicates of the study treatments (Table 2). Moreover, the

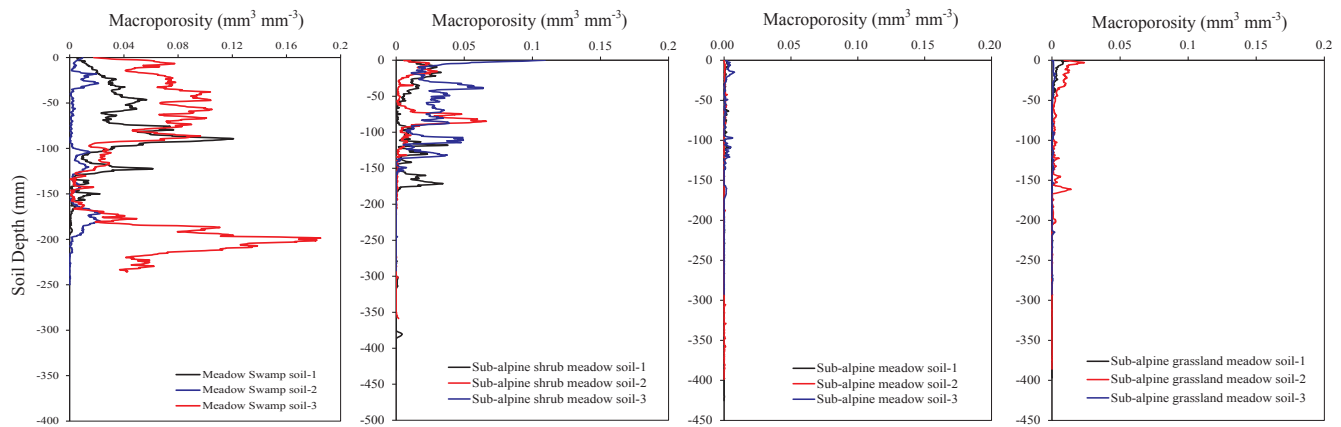


Fig. 4. Distribution of CT-measured macroporosity and SD along the soil column depth of the four soil types.

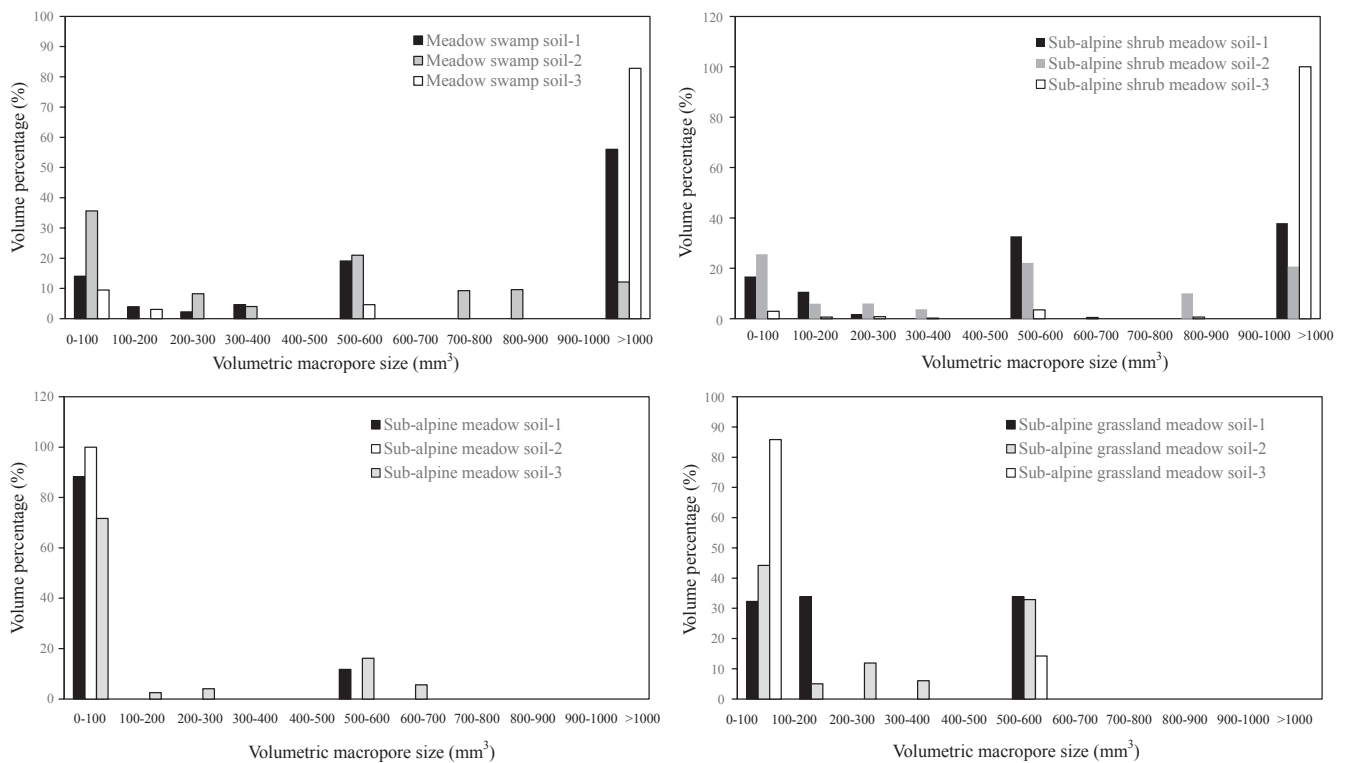


Fig. 5. Macropore size distribution (by volume) in the soil columns of the four soil types.

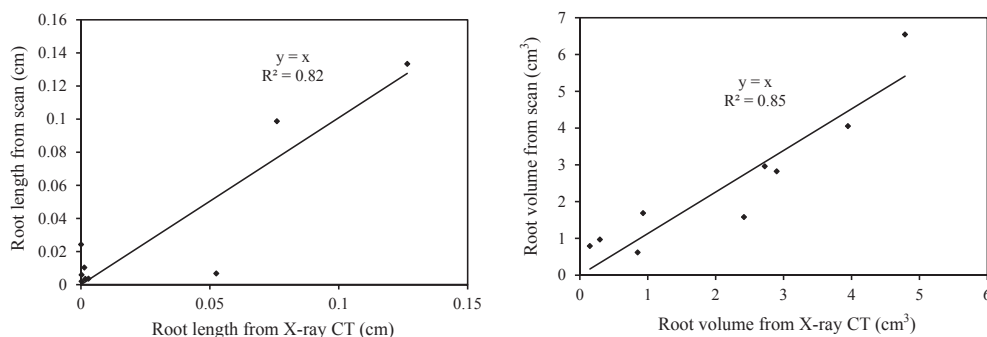


Fig. 6. Comparisons between the root length and root volume determined by X-ray CT and scanning after washing.

mean macropore sizes of *Meadow swamp* soil, *Sub-alpine shrub meadow* soil, *Sub-alpine meadow* soil and *Sub-alpine grassland meadow* soil were 5.4543 mm^3 , 5.3000 mm^3 , 1.7540 mm^3 and 1.8478 mm^3 , respectively.

The volume percentage of the macropores decreased and then increased as the macropore size increased for all typical meadow soil types (Fig. 5). The results indicated that, more than 50% of the

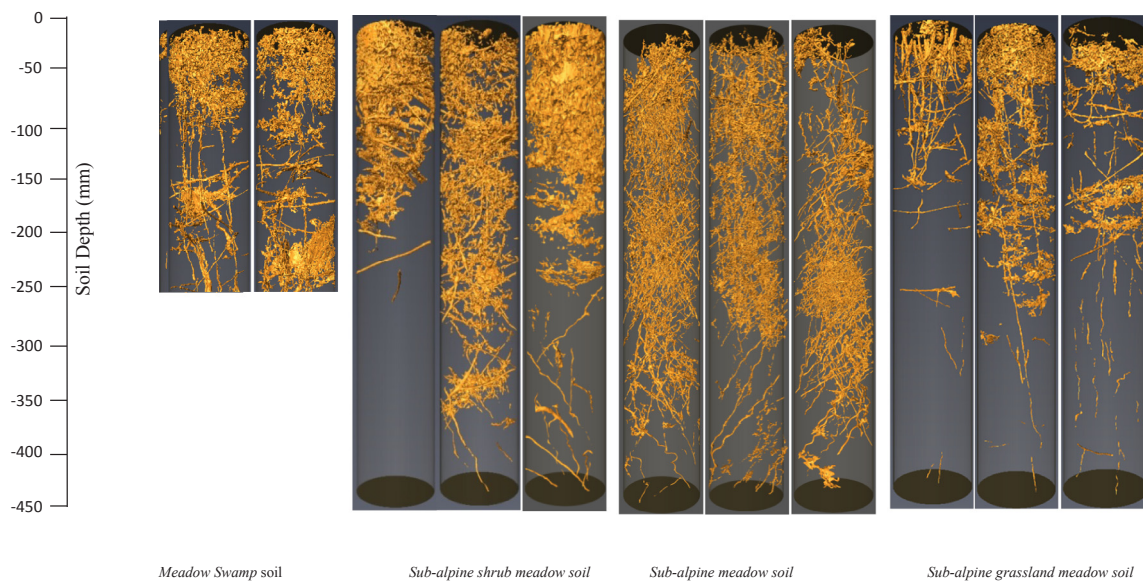


Fig. 7. Three-dimensional visualization of the root networks in the soil columns of the four soil types. Yellow denotes roots, and grey denotes non-pores. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

macropore volume consisted of macropores with a volume $> 1000 \text{ mm}^3$ for the *Meadow swamp* soil, *Sub-alpine shrub meadow* soil (50.32% and 52.90%), and more than 50% of the macropore volume consisted of macropores with a volume $< 100 \text{ mm}^3$ for *Sub-alpine meadow* soil (71.67%), and *Sub-alpine grassland meadow* soil (54.10%). Alpine meadow swamp soil mainly occurs in patches or strips in the mountains under waterlogged or moist soil conditions. These results demonstrate that the soil water environment and alpine shrub induce greater macroporosity of *Meadow swamp* soil and *Sub-alpine shrub meadow* soil than *Sub-alpine grassland meadow* soil and *Sub-alpine meadow* soil, and thus present a greater likelihood of larger macropores.

3.5. Visualization of root networks and root density

The correlations between the estimated root length and root volume determined by X-ray CT and those determined by the standard root washing method are shown in Fig. 6. There were good correlations between the root lengths, with a calculated coefficient of determination $R^2 = 0.82$; and root volumes, with $R^2 = 0.85$, determined by the two methods (Fig. 6). These results show that X-ray CT is a suitable method for root evaluation in situ due to its greater resolution of roots.

3-D visualizations of roots in the twelve soil columns are shown in Fig. 7. The spatial distribution of roots clearly differed among typical meadow soil types. The roots in the *Meadow swamp* soil and *Sub-alpine shrub meadow* soil were larger, deeper, and longer than those in the *Sub-alpine meadow* soil than *Sub-alpine grassland meadow* soil. More roots were concentrated in the deeper layer of the *Meadow swamp* soil and *Sub-alpine shrub meadow* soil than in those of the *Sub-alpine meadow* soil than *Sub-alpine grassland meadow* soil.

The root characteristics in the four treatments are listed in Table 3. The mean root density of the *Sub-alpine shrub meadow* soil was the highest ($0.0450 \text{ mm}^3 \text{ mm}^{-3}$), followed by that of the *Meadow swamp* soil ($0.0340 \text{ mm}^3 \text{ mm}^{-3}$), *Sub-alpine meadow* soil ($0.0023 \text{ mm}^3 \text{ mm}^{-3}$) and *Sub-alpine grassland meadow* soil ($0.0121 \text{ mm}^3 \text{ mm}^{-3}$) (Table 4). Significant differences were found in mean root density between *Sub-alpine shrub meadow* soil and *Meadow swamp* soil, *Sub-alpine meadow* soil, *Sub-alpine grassland meadow* soil.

The root surface area density of the *Sub-alpine shrub meadow* soil ($0.0785 \text{ m}^2 \text{ m}^{-3}$) was greater than those of the *Meadow swamp* soil ($0.0570 \text{ m}^2 \text{ m}^{-3}$), *Sub-alpine meadow* soil ($0.0499 \text{ m}^2 \text{ m}^{-3}$) and *Sub-alpine grassland meadow* soil ($0.0417 \text{ m}^2 \text{ m}^{-3}$). As expected, the mean root

length density and node density of the *Meadow swamp* soil and *Sub-alpine shrub meadow* soil ($30.4512 \times 10^3 \text{ m m}^{-3}$ and $75.8800 \times 10^5 \text{ m}^{-3}$, $30.1233 \times 10^3 \text{ m m}^{-3}$ and $59.0567 \times 10^5 \text{ m}^{-3}$ respectively) were higher than those of the *Sub-alpine meadow* soil ($21.6000 \times 10^3 \text{ m m}^{-3}$ and $50.6333 \times 10^5 \text{ m}^{-3}$, respectively), *Sub-alpine grassland meadow* soil ($13.9351 \times 10^3 \text{ m m}^{-3}$ and $37.1618 \times 10^5 \text{ m}^{-3}$, respectively). The root branch densities of the *Meadow swamp* soil and *Sub-alpine shrub meadow* soil were 55.2821 m^{-3} and 95.4433 m^{-3} , respectively. In addition, the root branch densities of the *Sub-alpine meadow* soil than *Sub-alpine grassland meadow* soil were 62.0967 m^{-3} and 55.2821 m^{-3} (Table 4).

3.6. Soil near saturated hydraulic conductivity

Of all the soil columns studied, *Meadow swamp* soil and *Sub-alpine shrub meadow* soil columns presented relatively continuous macropore networks, and the continuity of the *Sub-alpine meadow* soil than *Sub-alpine grassland meadow* soil columns was relatively low.

This study preliminarily reveals that a high continuity among soil macropores could increase the soil hydraulic conductivity in the *Meadow swamp* soil and *Sub-alpine shrub meadow* soil, emphasized the importance of soil macropore continuity and its connection to the soil hydrological behaviour in alpine environments. The soil near saturated hydraulic conductivity was significantly greater for the *Meadow swamp* soil ($K(10) = 2.274 \text{ mm} \cdot \text{h}^{-1}$ and $K(60) = 12.505 \text{ mm} \cdot \text{h}^{-1}$) and *Sub-alpine shrub meadow* soil ($K(10) = 0.885 \text{ mm} \cdot \text{h}^{-1}$ and $K(60) = 0.176 \text{ mm} \cdot \text{h}^{-1}$) than for the *Sub-alpine meadow* soil ($K(10) = 0.756 \text{ mm} \cdot \text{h}^{-1}$ and $K(60) = 0.078 \text{ mm} \cdot \text{h}^{-1}$) and *Sub-alpine grassland meadow* soil ($K(10) = 0.345 \text{ mm} \cdot \text{h}^{-1}$ and $K(60) = 0.059 \text{ mm} \cdot \text{h}^{-1}$) (Table 4).

As expected, the hydraulic radius of the *Sub-alpine shrub meadow* soil was the highest and presented a mean of 0.8167 mm , while the *Meadow swamp* soil, *Sub-alpine meadow* soil than *Sub-alpine grassland meadow* soil mean hydraulic radius were 0.6702 mm , 0.5548 mm and 0.5810 mm , respectively (Table 2). So, water could preferentially transport through macropores in *Meadow swamp* soil and *Sub-alpine shrub meadow* soil compared with *Sub-alpine meadow* soil than *Sub-alpine grassland meadow* soil.

Table 3
Root characteristics of all soil columns as determined using the CT images of the four soil types.

Soil types	Root density (mm ³ mm ⁻³)	Surface area density (m ² m ⁻³)	Length density (km m ⁻³)	Node density (×10 ⁵) (no.m ⁻³)	Branch density (no.m ⁻³)	Tortuosity (-)	Mean angle (°C)
<i>Meadow Swamp</i> soil	0.0340 (± 0.006)a	0.0570 (± 0.020)	30.4512 (± 7.15)a	75.8800 (± 12.93)	55.2821 (± 55.8195)	100.05 (± 22.07)	20.0324 (± 10.17)
<i>Sub-alpine shrub meadow</i> soil	0.0450 (± 0.015)b	0.0785 (± 0.01)	30.1233 (± 7.553)b	59.0567 (± 16.4761)	95.4433 (± 29.4494)	1.72 (± 0.6324)	51.4333 (± 3.8814)
<i>Sub-alpine meadow</i> soil	0.0023 (± 0.0002)a	0.0499 (± 0.0366)	21.6000 (± 4.0763)	50.6333 (± 12.411)	62.0967 (± 15.0043)	1.17 (± 0.01)	52.3767 (± 1.5054)
<i>Sub-alpine grassland meadow</i> soil	0.0121 (± 0.0059)a	0.0417 (± 0.0263)	13.9351 (± 10.6653)a	37.1618 (± 34.5845)	55.2821 (± 55.8195)	1.1758 (± 0.0203)	53.4389 (± 4.5601)

Data in the parentheses are standard deviations.

Table 4

Near saturation conductivity (K) (mm h⁻¹) of the four soil types.

Soil types	$K(10)$	$K(30)$	$K(60)$
<i>Meadow Swamp</i> soil	2.274	5.684	12.505
<i>Sub-alpine shrub meadow</i> soil	0.885	0.870	0.176
<i>Sub-alpine grassland meadow</i> soil	0.756	0.745	0.078
<i>Sub-alpine meadow</i> soil	0.345	0.300	0.059

3.7. Relationship between roots and soil macropores and near saturated hydraulic conductivity

Table 5 is the relationship between roots and soil macropores and near saturated hydraulic conductivity. The correlation analysis results presented in Table 5 show that root density was significantly and positively correlated with macroporosity, surface area density, length density, node density, branch density and hydraulic radius. Root length density was significantly and positively correlated with macroporosity, macropore surface area density, length density, node density, tortuosity and hydraulic radius. Root node density was significantly and positively correlated with macropore number density. Root surface area density was significantly and positively correlated with macropore length density, node density and branch density. And, root branch density was significantly and negatively correlated with soil macropore number density, length density, node density and branch density.

The correlation analysis results presented in Table 5 also show that soil macropore length density was significantly and positively correlated to soil near saturated hydraulic conductivity $K(10)$.

4. Discussion

The macropore networks were distinctly different among different soil types under alpine meadows. Different types of macropores were observed in the X-ray computed soil profile. Luo et al. (2010) reported that macropores formed by earthworms were highly continuous, relatively large, and tubular, and macropores formed by roots were highly continuous and round in shape, their size generally decreasing with depth. The smaller, more randomly distributed and less continuously distributed macropores were likely inter-aggregate macropores, such as those formed by freezing and thawing or wetting and drying (Luo et al., 2010). Therefore, the greater macroporosity and more tortuous paths in the *Meadow swamp* soil were closely associated with greater root development and subsequent root decay. And, the macroporosity in the *Sub-alpine shrub meadow* soil and *Sub-alpine grassland meadow* soil was associated with root development and freezing-thawing cycles. The smaller, more randomly distributed and less continuously distributed macropores in the *Sub-alpine meadow* soil were likely inter-aggregate macropores, such as those formed by freezing and thawing.

The results showed, the macroporosity of *Meadow swamp* soil and *Sub-alpine shrub meadow* soil was larger than that of *Sub-alpine grassland meadow* soil and *Sub-alpine meadow* soil. And, soil macropore length density was significantly and positively correlated to soil near saturated hydraulic conductivity $K(10)$. This is consistent with Borges et al. (2019) and Pan et al. (2017), K_s was directly influenced by macroporosity (Nakano and Miyazaki, 2005; Bhattacharyya et al., 2006). Soil moisture plays a critical role in land surface processes and hydrological cycles. It not only directly participates in soil hydrological processes but also influences vegetation growth and even modifies weather processes and local climates (Legates et al., 2011; Shein, 2010; Vereecken et al., 2015). So, higher soil water content in *Alpine meadow swamp* environment can promote vegetation root growth, which could induce soil macropore development. So, soil water environment and alpine shrub induce greater macroporosity of meadow soil, and thus present a greater likelihood of larger macropores.

The plant root system is critical in the development of soil

Table 5
Correlations between the soil macropore parameters and soil properties, root parameters, near saturation conductivity.

	Macropore number density ($\times 10^5$)	Macroporosity	Surface area density	Length density	Node density ($\times 10^5$)	Branch density	Tortuosity	Mean angle	Hydraulic radius	Mean macropore size	mean Coordination Number
Root density	0.243	0.909**	0.798**	0.876**	0.821**	0.805**	-0.401	0.158	0.907**	0.484	0.677*
Root number density	0.130	-0.008	0.003	0.065	0.103	0.098	0.000	-0.100	-0.115	-0.241	-0.056
Root surface area density	0.551	0.603	0.512	0.701*	0.716*	0.688*	-0.737*	0.158	0.554	0.267	0.595
Root length density	0.324	0.762*	0.715*	0.708*	0.676*	0.643	-0.181	0.318	0.694*	0.144	0.486
Root node density	0.669*	0.398	0.348	0.539	0.571	0.541	-0.127	0.293	0.261	-0.077	0.341
Root branch density	0.702*	0.580	0.516	0.707*	0.729*	0.692*	-0.254	0.254	0.455	-0.007	0.504
Root tortuosity	-0.058	0.406	0.250	0.406	0.341	0.364	-0.376	0.195	0.528	0.864**	0.261
Root mean angle	0.549	-0.453	-0.538	-0.109	-0.017	-0.006	-0.082	0.477	-0.468	0.123	-0.106
K(10)	0.351	0.743	0.570	0.852*	0.806	0.811	-0.302	0.127	0.721	0.568	0.611
K(30)	0.250	0.771	0.632	0.782	0.727	0.714	-0.545	-0.233	0.741	0.479	0.461
K(60)	0.314	0.610	0.406	0.769	0.727	0.748	-0.401	0.286	0.667	0.793	0.648

macropores via root decay and penetration to depths in the soil. The below-ground biomass of plants, which is a major component of total primary production, plays an important role in the productivity of plants and the storage of vegetation C (Wu et al., 2011). The root system serves as a dynamic interface between soils and plants (Cai and Shen, 2002) and acts as an important structural and functional component of ecosystems. The results showed that, root density was significantly and positively correlated with macroporosity, soil macropore surface area density, length density, node density, branch density and hydraulic radius. Root node density was significantly and positively correlated with macropore number density. Root length density was significantly and positively correlated with macroporosity, macropore surface area density, length density, node density tortuosity and hydraulic radius. Root surface area density was significantly and positively correlated with macropore length density, node density and branch density. And, root branch density was significantly and positively correlated with soil macropore number density, length density, node density and branch density. So, root density, root node density, root length density, root surface area density and branch density were important root parameters influencing soil macropore, which were the connectivity parameters influencing soil macropore development. As plant roots could enhance soil aggregation, thus forming inter-aggregate macropores, serving as preferential flow paths (Fageria and Stone, 2006; Ghestem et al., 2011). Additionally, root channels could create well-connected networks that contribute to water percolation (Noguchi et al., 1997; Sidle et al., 2001; Newman et al., 2004). So, root connectivity parameters (root density, root node density, root length density, root surface area density and branch density) were important factors influencing soil macropore development.

5. Conclusions

This study quantified the soil macropore and root architecture of the Meadow swamp soil, Sub-alpine shrub meadow soil, Sub-alpine grassland meadow soil and Sub-alpine meadow soil of the Qinghai Lake watershed using CT imaging. The results showed that, the soil macropore networks were distinctly different among different soil types under alpine meadows. More macropores of Sub-alpine grassland meadow soil and Sub-alpine meadow soil were distributed in the surface layer than in the deeper layers of the Meadow swamp soil and Sub-alpine shrub meadow soil, while macropores were predominantly distributed throughout the 0–400-mm soil layer in the Sub-alpine shrub meadow soil. Moreover, the soil macropores of the Meadow swamp soil and Sub-alpine shrub meadow soil were more convoluted and continuous than those of the Sub-alpine grassland meadow soil and Sub-alpine meadow soil. Root density was greater in the Meadow swamp soil and Sub-alpine shrub meadow soil than those in the Sub-alpine grassland meadow soil and Sub-alpine meadow soil. The macropore architecture could facilitate greater water flow to deeper soil layers in the Meadow swamp soil and Sub-alpine shrub meadow soil than in the Sub-alpine grassland meadow soil and Sub-alpine meadow soil. Root density was significantly and positively correlated with macroporosity, soil macropore surface area density, length density, node density, branch density and hydraulic radius. Root node density was significantly and positively correlated with macropore number density. And, root branch density was significantly and positively correlated with soil macropore number density, length density, node density and branch density. And, soil macropore length density was significantly and positively correlated to soil near saturated hydraulic conductivity $K(10)$. So, soil water environment and alpine shrub induce greater macroporosity of meadow soil, and thus present a greater likelihood of larger macropores. Root connectivity parameters (root density, root node density, root length density, root surface area density and branch density) were important parameters influencing soil macropore development.

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